Chapter 6 : L^p Spaces

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Chapter 6 : L^p Spaces

Problem 6.1. When does equality hold in Minkowski's inequality? (The answer is different for $p = 1$ and for $1 < p < \infty$. What about $p = \infty$?)

Problem 6.2. Prove Theorem 6.8.

Problem 6.3. If $1 \le p < r \le \infty$, $L^p \cap L^r$ is a Banach space with norm $\|f\| = \|f\|_p + \|f\|_r$, and if $p < q < r$, the inclusion map $L^p \cap L^r \to L^q$ is continuous.

Problem 6.4. If $1 \le p < r \le \infty$, $L^p + L^r$ is a Banach space with norm $\|f\| = \inf\{\|g\|_p + \|h\|_r : f = g + h\}$, and if $p < q < r$, the inclusion map $L^q \to L^p + L^r$ is continuous.

Problem 6.5. Suppose $0 < p < q < \infty$. Then $L^p \not\subset L^q$ iff X contains sets of arbitrarily small positive measure, and $L^q \not\subset L^p$ iff X contains sets of arbitrarily large finite measure. (For the "if" implication: In the first case there is a disjoint sequence $\{E_n\}$ with $0 < \mu(E_n) < 2^{-n}$, and in the second case there is a disjoint sequence $\{E_n\}$ with $1 \le \mu(E_n) < \infty$. Consider $f = \sum a_n \chi_{E_n}$ for suitable constants a_n.) What about the case $q = \infty$?

Problem 6.6. Suppose $0 < p_0 < p_1 \le \infty$. Find examples of functions f on $(0, \infty)$ (with Lebesgue measure), such that $f \in L^p$ iff (a) $p_0 < p < p_1$, (b) $p_0 \le p \le p_1$, (c) $p = p_0$. (Consider functions of the form $f(x) = x^{-a} |\log x|^b$.)

Problem 6.7. If $f \in L^p \cap L^\infty$ for some $p < \infty$, so that $f \in L^q$ for all $q > p$, then $\|f\|_\infty = \lim_{q\to\infty} \|f\|_q$.

Problem 6.8. Suppose $\mu(X) = 1$ and $f \in L^p$ for some $p > 0$, so that $f \in L^q$ for $0 < q < p$.

a. $\log \|f\|_q \ge \int \log |f|$. (Use Exercise 42d in §3.5, with $F(t) = e^t$.)

b. $(\int |f|^q - 1)/q \ge \log \|f\|_q$ and $(\int |f|^q - 1)/q \to \int \log |f|$ as $q \to 0$.

c. $\lim_{q\to 0} \|f\|_q = \exp\Big(\int \log |f|\Big)$.

Problem 6.9. Suppose $1 \le p < \infty$. If $\|f_n - f\|_p \to 0$ then $f_n \to f$ in measure, and hence some subsequence converges to f a.e. On the other hand, if $f_n \to f$ in measure and $|f_n| \le g \in L^p$ for all n, then $\|f_n - f\|_p \to 0$.

Problem 6.10. Suppose $1 \le p < \infty$. If $f_n, f \in L^p$ and $f_n \to f$ a.e., then $\|f_n - f\|_p \to 0$ iff $\|f_n\|_p \to \|f\|_p$. (Use Exercise 20 in §2.3.)

Problem 6.11. If f is a measurable function on X, define the essential range R_f of f to be the set of all $z \in \mathbb{C}$ such that $\{x : |f(x) - z| < \epsilon\}$ has positive measure for all $\epsilon > 0$.

a. R_f is closed.

b. If $f \in L^\infty$ then R_f is compact and $\|f\|_\infty = \max\{|z| : z \in R_f\}$.

Problem 6.12. If $p \neq 2$, the L^p norm does not arise from an inner product on L^p, except in trivial cases when $\dim(L^p) \leq 1$. (Show that the parallelogram law fails.)

Problem 6.13. $L^p(\mathbb{R}^n, m)$ is separable for $1 \leq p < \infty$. However, $L^\infty(\mathbb{R}^n, m)$ is not separable. (There is an uncountable set $\mathcal{F} \subset L^\infty$ such that $\|f - g\|_\infty \geq 1$ for all $f, g \in \mathcal{F}$ with $f \neq g$.)

Problem 6.14. If $g \in L^\infty$ the operator T defined by $Tf = fg$ is bounded on L^p for $1 \leq p \leq \infty$. Its operator norm is at most $\|g\|_\infty$, with equality if μ is semifinite.

Problem 6.15. (The Vitali Convergence Theorem) Suppose $1 \leq p < \infty$ and $\{f_n\}_1^\infty \subset L^p$. In order for $\{f_n\}$ to be Cauchy in the L^p norm it is necessary and sufficient for the following three conditions to hold: (i) $\{f_n\}$ is Cauchy in measure; (ii) the sequence $\{|f_n|^p\}$ is uniformly integrable (see Exercise 11 in §3.2); and (iii) for every $\epsilon > 0$ there exists $E \subset X$ such that $\mu(E) < \infty$ and $\int_{E^c} |f_n|^p < \epsilon$ for all n. (To prove the sufficiency: Given $\epsilon > 0$, let E be as in (iii), and let $A_{mn} = \{x \in E : |f_m(x) - f_n(x)| \geq \epsilon\}$. Then the integrals of $|f_n - f_m|^p$ over $E \backslash A_{mn}$, A_{mn}, and E^c are small when m and n are large for three different reasons.)

Problem 6.16. If $0 < p < 1$, the formula $\rho(f, g) = \int |f - g|^p$ defines a metric on L^p that makes L^p into a complete topological vector space. (The proof of Theorem 6.6 still works for $p < 1$ if $\|f\|_p$ is replaced by $\int |f|^p$, as it uses only the triangle inequality and not the homogeneity of the norm.)

Problem 6.17. With notation as in Theorem 6.14, if μ is semifinite, $q < \infty$, and $M_q(g) < \infty$, then $\{x : |g(x)| > \epsilon\}$ has finite measure for all $\epsilon > 0$ and hence S_g is σ-finite.

Problem 6.18. The self-duality of L^2 follows from Hilbert space theory (Theorem 5.25), and this fact can be used to prove the Lebesgue-Radon-Nikodym theorem by the following argument due to von Neumann. Suppose that μ, ν are positive finite measures on $(X, \mathcal{M})$ (the σ-finite case follows easily as in §3.2), and let $\lambda = \mu + \nu$.

a. The map $f \mapsto \int f \, d\nu$ is a bounded linear functional on $L^2(\lambda)$, so $\int f \, d\nu =$

$\int fg\,d\lambda$ for some $g \in L^2(\lambda)$. Equivalently, $\int f(1-g)\,d\nu = \int fg\,d\mu$ for $f \in L^2(\lambda)$.

b. $0 \le g \le 1$ λ-a.e., so we may assume $0 \le g \le 1$ everywhere.

c. Let $A = \{x : g(x) < 1\}$, $B = \{x : g(x) = 1\}$, and set $\nu_a(E) = \nu(A \cap E)$, $\nu_s(E) = \nu(B \cap E)$. Then $\nu_s \perp \mu$ and $\nu_a \ll \mu$; in fact, $d\nu_a = g(1-g)^{-1}\chi_A\,d\mu$.

Problem 6.19. Define $\phi_n \in (l^\infty)^*$ by $\phi_n(f) = n^{-1}\sum_1^n f(j)$. Then the sequence $\{\phi_n\}$ has a weak* cluster point ϕ, and ϕ is an element of $(l^\infty)^*$ that does not arise from an element of l^1.

Problem 6.20. Suppose $\sup_n \|f_n\|_p < \infty$ and $f_n \to f$ a.e.

a. If $1 < p < \infty$ then $f_n \to f$ weakly in L^p. (Given $g \in L^q$, where q is conjugate to p, and $\epsilon > 0$, there exist (i) $\delta > 0$ such that $\int_E |g|^q < \epsilon$ whenever $\mu(E) < \delta$, (ii) $A \subset X$ such that $\mu(A) < \infty$ and $\int_{X\setminus A} |g|^q < \epsilon$ and (iii) $B \subset A$ such that $\mu(A\setminus B) < \delta$ and $f_n \to f$ uniformly on B.)

b. The result of (a) is false in general for $p = 1$. (Find counterexamples in $L^1(\mathbb{R}, m)$ and l^1.) It is, however, true for $p = \infty$ if μ is σ-finite and weak convergence is replaced by weak* convergence.

Problem 6.21. If $1 < p < \infty$, $f_n \to f$ weakly in $l^p(A)$ iff $\sup_n \|f_n\|_p < \infty$ and $f_n \to f$ pointwise.

Problem 6.22. Let $X = [0, 1]$, with Lebesgue measure.

a. Let $f_n(x) = \cos 2\pi nx$. Then $f_n \to 0$ weakly in L^2 (see Exercise 63 in §5.5), but $f_n \not\to 0$ a.e. or in measure.

b. Let $f_n(x) = n\chi_{(0,1/n)}$. Then $f_n \to 0$ a.e. and in measure, but $f_n \not\to 0$ weakly in L^p for any p.

Problem 6.23. Let $(X, \mathcal{M}, \mu)$ be a measure space. A set $E \in \mathcal{M}$ is called locally null if $\mu(E \cap F) = 0$ for every $F \in \mathcal{M}$ such that $\mu(F) < \infty$. If $f : X \to \mathbb{C}$ is a measurable function, define $\|f\|_* = \inf\{a : \{x : |f(x)| > a\}$ is locally null$\}$, and let $\mathcal{L}^\infty = \mathcal{L}^\infty(X, \mathcal{M}, \mu)$ be the space of all measurable f such that $\|f\|_* < \infty$. We consider $f, g \in \mathcal{L}^\infty$ to be identical if $\{x : f(x) \ne g(x)\}$ is locally null.

a. If E is locally null, then $\mu(E)$ is either 0 or ∞. If μ is semifinite, then every locally null set is null.

b. $\|\cdot\|_*$ is a norm on $\mathcal{L}^\infty$ that makes $\mathcal{L}^\infty$ into a Banach space. If μ is semifinite, then $\mathcal{L}^\infty = L^\infty$.

Problem 6.24. If $g \in \mathcal{L}^\infty$ (see Exercise 23), then $\|g\|_* = \sup\{|\int fg| : \|f\|_1 = 1\}$, so the map $g \mapsto \phi_g$ is an isometry from $\mathcal{L}^\infty$ into $(L^1)^*$. Conversely, if $M_\infty(g) < \infty$ as in Theorem 6.14, then $g \in \mathcal{L}^\infty$ and $M_\infty(g) = \|g\|_*$.

Problem 6.25. Suppose μ is decomposable (see Exercise 15 in §3.2). Then every $\phi \in (L^1)^*$ is of the form $\phi(f) = \int fg$ for some $g \in \mathcal{L}^\infty$, and hence $(L^1)^* \cong \mathcal{L}^\infty$ (see Exercises 23 and 24). (If $\mathcal{F}$ is a decomposition of μ and $f \in L^1$, there exists $\{E_j\} \subset \mathcal{F}$ such that $f = \sum_1^\infty f\chi_{E_j}$ where the series converges in L^1.)

Problem 6.26. Complete the proof of Theorem 6.18 for the cases $p = 1$ and $p = \infty$.

Problem 6.27. (Hilbert's Inequality) The operator $Tf(x) = \int_0^\infty (x+y)^{-1} f(y)\,dy$ satisfies $\|Tf\|_p \le C_p\|f\|_p$ for $1 < p < \infty$, where $C_p = \int_0^\infty x^{-1/p}(x+1)^{-1}\,dx$. (For those who know about contour integrals: Show that $C_p = \pi \csc(\pi/p)$.)

Problem 6.28. Let I_α be the αth fractional integral operator as in Exercise 61 of §2.6, and let $J_\alpha f(x) = x^{-\alpha} I_\alpha f(x)$.

a. J_α is bounded on $L^p(0,\infty)$ for $1 < p \le \infty$; more precisely, $\|J_\alpha f\|_p \le \frac{\Gamma(1-p^{-1})}{\Gamma(\alpha+1-p^{-1})}\|f\|_p$.

b. There exists $f \in L^1(0,\infty)$ such that $J_1 f \notin L^1(0,\infty)$.

Problem 6.29. Suppose that $1 \le p < \infty$, $r > 0$, and h is a nonnegative measurable function on $(0,\infty)$. Then: $\int_0^\infty x^{-r-1}\left[\int_0^x h(y)\,dy\right]^p dx \le \left(\frac{p}{r}\right)^p \int_0^\infty x^{p-r-1}h(x)^p\,dx$, $\int_0^\infty x^{r-1}\left[\int_x^\infty h(y)\,dy\right]^p dx \le \left(\frac{p}{r}\right)^p \int_0^\infty x^{p+r-1}h(x)^p\,dx$. (Apply Theorem 6.20 with $K(x,y) = x^{\beta-1}y^{-\beta}\chi_{(0,\infty)}(y-x)$, $f(x) = x^\gamma h(x)$, and $g(x) = x^\delta h(x)$ for suitable β, γ, δ.)

Problem 6.30. Suppose that K is a nonnegative measurable function on $(0,\infty)$ such that $\int_0^\infty K(x)x^{s-1}\,dx = \phi(s) < \infty$ for $0 < s < 1$.

a. If $1 < p < \infty$, $p^{-1} + q^{-1} = 1$, and f, g are nonnegative measurable functions on $(0,\infty)$, then (with $\int = \int_0^\infty$) $\iint K(xy)f(x)g(y)\,dx\,dy \le \phi(p^{-1})\left[\int x^{p-2}f(x)^p\,dx\right]^{1/p}\left[\int g(x)^q\,d\right.$

b. The operator $Tf(x) = \int_0^\infty K(xy)f(y)\,dy$ is bounded on $L^2((0,\infty))$ with norm $\le \phi(\frac{1}{2})$. (Interesting special case: If $K(x) = e^{-x}$ then T is the Laplace transform and $\phi(s) = \Gamma(s)$.)

Problem 6.31. (A Generalized Hölder Inequality) Suppose that $1 \le p_j \le \infty$ and $\sum_1^n p_j^{-1} = r^{-1} \le 1$. If $f_j \in L^{p_j}$ for $j = 1, \dots, n$ then $\prod_1^n f_j \in L^r$ and $\|\prod_1^n f_j\|_r \le \prod_1^n \|f_j\|_{p_j}$. (First do the case $n = 2$.)

Problem 6.32. Suppose that $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are σ-finite measure spaces and $K \in L^2(\mu \times \nu)$. If $f \in L^2(\nu)$, the integral $Tf(x) = \int K(x,y) f(y)\, d\nu(y)$ converges absolutely for a.e. $x \in X$ moreover, $Tf \in L^2(\mu)$ and $\|Tf\|_2 \le \|K\|_2 \|f\|_2$.

Problem 6.33. Given $1 < p < \infty$, let $Tf(x) = x^{-1/p} \int_0^x f(t)\, dt$. If $p^{-1} + q^{-1} = 1$ then T is a bounded linear map from $L^q((0,\infty))$ to $C_0((0,\infty))$.

Problem 6.34. If f is absolutely continuous on $[\epsilon, 1]$ for $0 < \epsilon < 1$ and $\int_0^1 x|f'(x)|^p\, dx < \infty$, then $\lim_{x\to 0} f(x)$ exists (and is finite) if $p > 2$, $|f(x)|/|\log x|^{1/2} \to 0$ as $x \to 0$ if $p = 2$, and $|f(x)|/x^{1-(2/p)} \to 0$ as $x \to 0$ if $p < 2$.

Problem 6.35. For any measurable f and g we have $[cf]_p = |c|[f]_p$ and $[f + g]_p \le 2([f]_p^p + [g]_p^p)^{1/p}$; hence weak L^p is a vector space. Moreover, the "balls" $\{g : [g - f]_p < r\}$ ($r > 0$, $f \in$ weak L^p) generate a topology on weak L^p that makes weak L^p into a topological vector space.

Problem 6.36. If $f \in$ weak L^p and $\mu(\{x : f(x) \ne 0\}) < \infty$, then $f \in L^q$ for all $q < p$. On the other hand, if $f \in$ (weak L^p) $\cap\, L^\infty$, then $f \in L^q$ for all $q > p$.

Problem 6.37. Prove Proposition 6.25.

Problem 6.38. $f \in L^p$ iff $\sum_{-\infty}^{\infty} 2^{kp} \lambda_f(2^k) < \infty$.

Problem 6.39. If $f \in L^p$, then $\lim_{\alpha\to 0} \alpha^p \lambda_f(\alpha) = \lim_{\alpha\to\infty} \alpha^p \lambda_f(\alpha) = 0$ (First suppose f is simple.)

Problem 6.40. If f is a measurable function on X, its decreasing rearrangement is the function $f^* : (0,\infty) \to [0,\infty]$ defined by $f^*(t) = \inf\{\alpha : \lambda_f(\alpha) \le t\}$ (where $\inf \emptyset = \infty$).

a. f^* is decreasing. If $f^*(t) < \infty$ then $\lambda_f(f^*(t)) \le t$, and if $\lambda_f(\alpha) < \infty$ then $f^*(\lambda_f(\alpha)) \le \alpha$.

b. $\lambda_f = \lambda_{f^*}$, where λ_{f^*} is defined with respect to Lebesgue measure on $(0,\infty)$.

c. If $\lambda_f(\alpha) < \infty$ for all $\alpha > 0$ and $\lim_{\alpha\to\infty} \lambda_f(\alpha) = 0$ (so that $f^*(t) < \infty$ for all $t > 0$), and ϕ is a nonnegative measurable function on $(0,\infty)$, then $\int_X \phi \circ |f|\, d\mu = \int_0^\infty \phi \circ f^*(t)\, dt$. In particular, $\|f\|_p = \|f^*\|_p$ for $0 < p < \infty$.

d. If $0 < p < \infty$, $[f]_p = \sup_{t>0} t^{1/p} f^*(t)$.

e. The name "rearrangement" for f^* comes from the case where f is a nonnegative function on $(0,\infty)$. To see why it is appropriate, pick a step function on

$(0, \infty)$ assuming four or five different values and draw the graphs of f and f^*.

Problem 6.41. Suppose $1 < p \leq \infty$ and $p^{-1} + q^{-1} = 1$. If T is a bounded operator on L^p such that $\int (Tf)g = \int f(Tg)$ for all $f, g \in L^p \cap L^q$, then T extends uniquely to a bounded operator on L^r for all r in $[p, q]$ (if $p < q$) or $[q, p]$ (if $q < p$).

Problem 6.42. Prove the Marcinkiewicz theorem in the case $p_0 = p_1$. (Setting $p = p_0 = p_1$, we have $\lambda_{Tf}(\alpha) \leq (C_0\|f\|_p/\alpha)^{q_0}$ and $\lambda_{Tf}(\alpha) \leq (C_1\|f\|_p/\alpha)^{q_1}$. Use whichever estimate is better, depending on α, to majorize $q\int_0^\infty \alpha^{q-1}\lambda_{Tf}(\alpha)\,d\alpha$.)

Problem 6.43. Let H be the Hardy-Littlewood maximal operator on $\mathbb{R}$. Compute $H\chi_{(0,1)}$ explicitly. Show that it is in L^p for all $p > 1$ and in weak L^1 but not in L^1, and that its L^p norm tends to ∞ like $(p-1)^{-1}$ as $p \to 1$, although $\|\chi_{(0,1)}\|_p = 1$ for all p.

Problem 6.44. Let I_α be the fractional integration operator of Exercise 61 in §2.6. If $0 < \alpha < 1$, $1 < p < \alpha^{-1}$ and $r^{-1} = p^{-1} - \alpha$, then I_α is weak type $(1, (1-\alpha)^{-1})$ and strong type (p, r) with respect to Lebesgue measure on $(0, \infty)$.

Problem 6.45. If $0 < \alpha < n$ define an operator T_α on functions on $\mathbb{R}^n$ by $T_\alpha f(x) = \int |x-y|^{-\alpha} f(y)\,dy$. Then T_α is weak type $(1, (n-\alpha)^{-1})$ and strong type (p, r) with respect to Lebesgue measure on $\mathbb{R}^n$, where $1 < p < n\alpha^{-1}$ and $r^{-1} = p^{-1} - \alpha n^{-1}$. (The case $n = 3$, $\alpha = 1$ is of particular interest in physics: If f represents the density of a mass or charge distribution, $-(4\pi)^{-1}T_1 f$ represents the induced gravitational or electrostatic potential.)